MIL-T-48460B(AR)
30 June 1981
SUPERSEDING
MiL-T-48460A(AR)
1 March 1976

MILITARy SPECIFICATION

TACHOMETER, RATE, COMPUTER: 11732700

This specificaricn is approved for use by the US Army Armament Research and Development Command (ARRRADCOM) and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Tachometer, Nate, computer: 11732700 which is a unit of the M21 Ballistic Computer, 11732979.

2. APPLICABLE

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

MILITARY

MIL-F-!3926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-I-45607	InsPeCtion Equipment , Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibrationn Systems Requirements

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-81O	Environmental Test Methods

DRAWINGS

ARRADCOM

11732700	Tachometer, Rate, Computer
11732978	Tachometer, Rate, Computer, Interface Control

Inspection Equipment

11750201	Test Set - Rate Tachometer
11821748	Test Set - Immersion Leak, Purge and Fill
11821411	Fixture, Rate Tachometer
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Adapter Plate, Shock/Vibration
11821461	Unit Fixture, Rate Tachometer
11833718	Rate Tachometer - Vibration Input Drive
11823404	Table, Rate of Turn
11821668	Tach Generator Drive Fixture - Altitude

Packaging Data Sheet

Pl1732700	Packaging of Tachometer, Rate, Computer: 11732700

(Copies of specifications, standards, drawings ad publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Computer Rate Tachometer, herein referred to as the assembly, shall be manufactured in accordance with Drawing 1132700 and drawings pertaining thereto and when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall probide a direct current voltage output which is proportional to the angular velocity of the turret.

3.1.2 General specifications. The following previsions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a. Temperature	73° + 18°F
b. Relative humidity	50 perccnt ± 30 percent
c. Atmospheric pressure	28.5 + 2.0 - 3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 Output volatage linearity. With the output load in figure 1 and in put conditions in table 1 applied, the output voltage linaearlty shall be as specified in table 1. Table 1 limits in column A shall apply for all ten items and limits per column B shall apply for not less than seven items. (See 4.6.2.1)

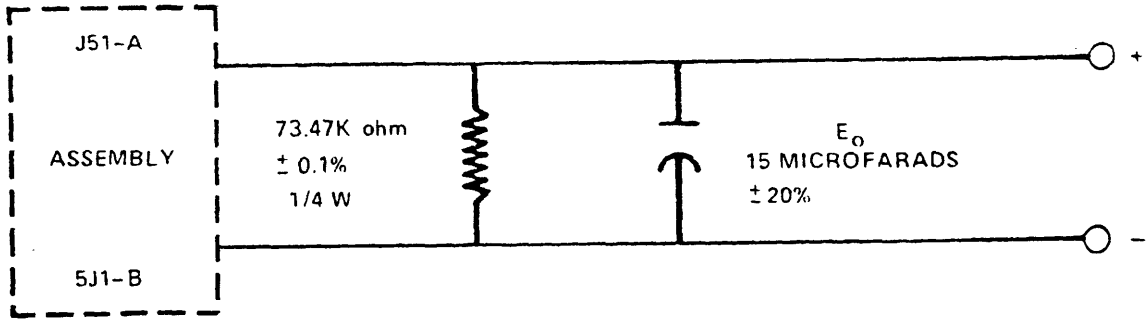


FIGURE 1. Output load accuracy measurement

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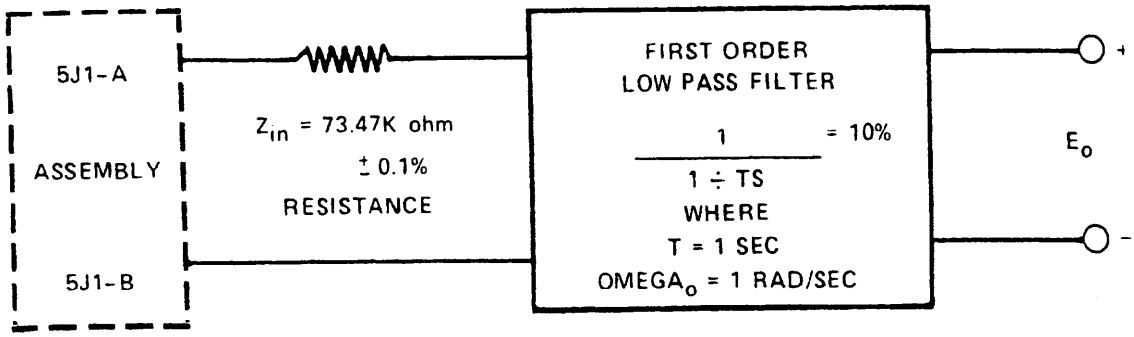


FIGURE 2. Output load for ripple measurement.

3.3.2 Output voltage ripple. With the output load in figure 2 and input conditions in table 1 applied, the output voltage ripple shall be as specified in table 1. (see 2.3.2.2)

TABLE I. Output voltage accuracy.

Item	Input Shaft Rate Milliradians/Sec	Output Voltage(E_o) Millivolts (mv)	Linearity Tolerance mv		% Ripple Tolerance p-p
			A	B	
1	1473	1473	20	10	2
2	589	589	10	5	-
3	295	295	6	3	3
4	147	147	4	2	-
5	59	59	2	1	6
6	-1473	-1473	20	10	2
7	-589	-589	10	5	-
8	-295	-295	6	3	3
9	-147	-147	4	2	-
10	-59	-59	2	1	6

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3.3.3 Rest output. With the assembly input shaft at rest, the output voltage shall be ± 0.5 mV (see 4.6.2.3)

3.3.4 Starting torque. The gear train of the assembly shall not bind when a starting torque of two inch-pounds is applied to the shaft end of the unit. (see 4.6.2.4)

3.4 Scaling. With an internal nitrogen pressure of 5.0 ± 0.1 psig, the assembly shall exhibit a pressure drop not greater than 0.2 psig for not less than 2 hours. The sealing og the assembly shall meet this criteria when all manual controls are exerciesed. The pressurized assembly shall be purged with dry nitrogen in accordance with BB-N-411, type I, grade B, class 1, at a dewpoint of -25°F of lower. The final internal nitrogen pressure shall be 0.0 + 0.2 - 0.0 psig. (see 4.6.4)

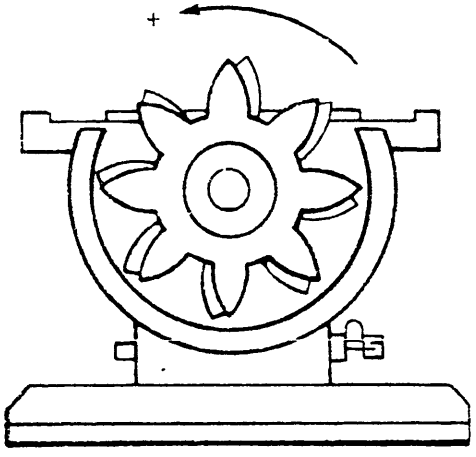


FIGURE 3. Positive shaft rotation.

3.5 Environmental.

3.5.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 50 ± 5 gravity units (g) for a duration of 18 ±3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 125 ± 12.5 g for a duration of 0.6 ± 0.2 msec applied in each dirction along three mutually perpendicular axes. (See 4.6.3.1)

3.5.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 4. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

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3.5.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.5.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.5.5 Humidity. The assembly shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.3.6)

3.5.6 Altitude. The assembly shall operate as specified herein under the following conditions:

a. during exzposure to an atmospheric pressure equivalent to 10000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (see 4.6.3.7)

3.5.7 Dust. The assembly shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (see 4.6.3.8)

3.5.8 Slat fog. The assembly shall operate in specified herein after exposure to 5 percent sodium chloride atomized spray for not less than 48 hours. (see 4.6.3.9)

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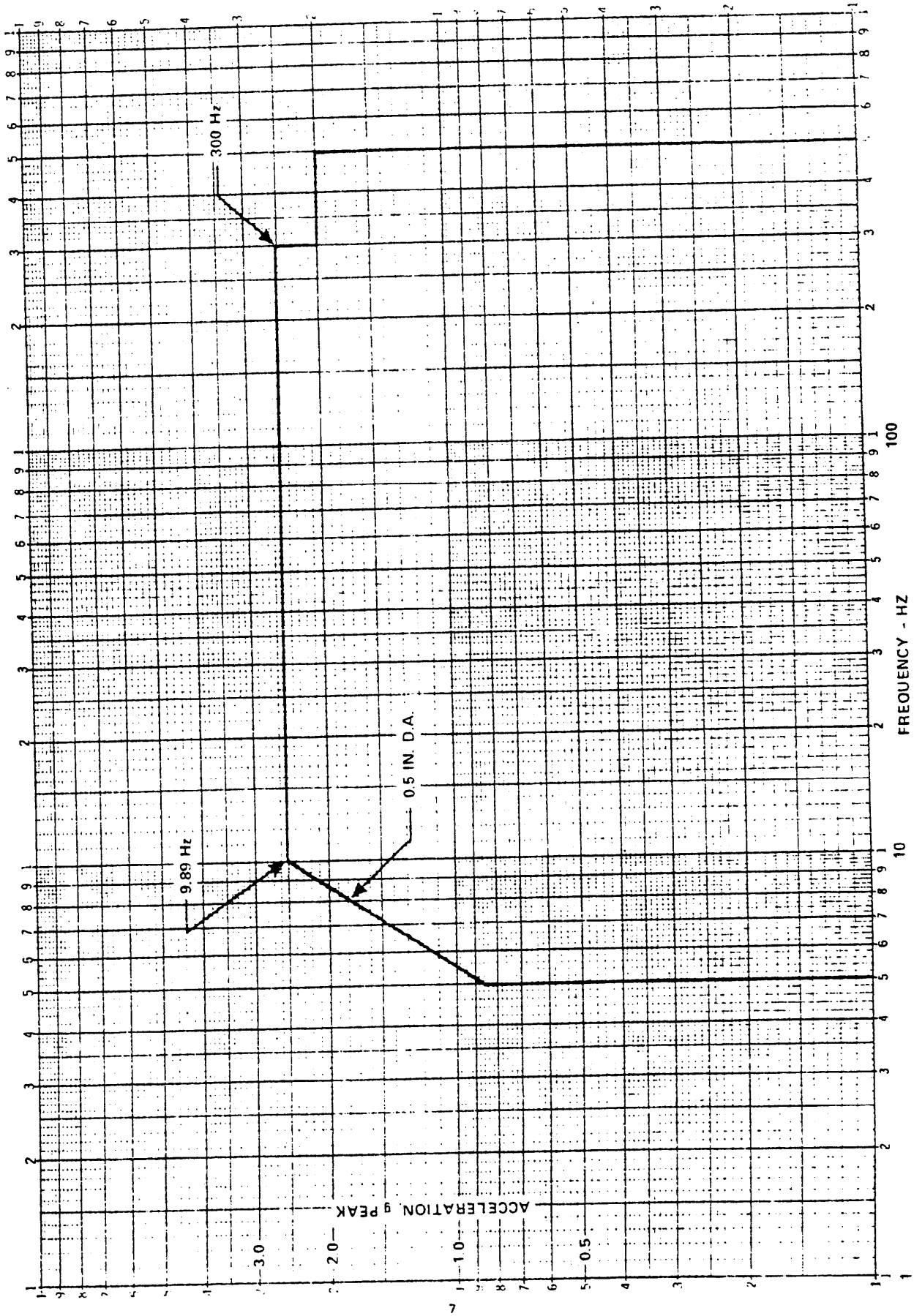


FIGURE 4. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance os the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services confom to prescribed requirements .

4.2 Classifications of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirements for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in table II.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requiraements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE II. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Fabrication	3.1	Applicable drawings - Visual
2	General Specification	3.1.2	MIL-F-13926 - Visual
3	Output Voltage Linearity	3.3.1	4.6.2.1
4	Output Voltage Ripple	3.3.2	4.6.2.2
5	Rest Output	3.3.3	4.6.2.3
6	Starting Torque	3.3.4	4.6.2.4
7	Sealing A and B	3.4	4.6.4
8	High Temperature	3.5.3 and 3.5.4	4.6.3.3
9	Low Temperature	3.5.3 and 3.5.4	4.6.3.4
10	Humidity	3.5.5	4.6.3.6
11	Altitude	3.5.6	4.6.3.2
12	Shock	3.5.1	4.6.3.1
13	Vibration A	3.5.2	4.6.3.2
14	Dust	3.5.7	4.6.3.8
15	Salt Fog	3.5.8	4.6.3.9
16	Temperature cycling	3.5.3 and 3.5.4	4.6.3.5

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4.5 Quality conformance inspection.

4. 5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table IV.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table V shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirements the lot shall be rejected. Rejected lots shall be subject to the provisions of MI.L-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table IV and only the test of item 307 in table V shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table 111 and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

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4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shll be in accordance with the provisions of packaging data sheet P11732700.

TABLE III. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Output Voltage Linearity	3.3.1	4.6.2.1
102	Outut Voltage Ripple	3.3.2	4.6.2.2
103	Rest Output	3.3.3	4.6.2.3
104	Starting Torque	3.3.4	4.6.2.4
105	Sealing A	3.4	4.6.4

TABLE IV. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

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TABLE V. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	High Temperature	3.5.3 and 3.5.4	4.6.3.3
304	Low Temperature	3.5.3 and 3.5.4	4.6.3.4
305	Vibration B	3.5.2	4.6.3.2
306	Sealing B	3.4	4.6.4
307	Temperature cycling	3.5.3 and 3.5.4	4.6.3.5

4.6 Methods of inspection.

4.6.1 Test equipment. The test wquipment in table VI shall be used to perfom the required tests where specified.

TABLE VI. Test equipment

Item	Part Number	Nomenclature
1	11750201	Test Set - Rate Tachometer
2	11821748	Test Set - Immersion Leak, Purge and Fill
3	11821411	Fixture, Rate Tachometer
4	11821457	Fixture Shock, Vertical and Horizontal Plane
5	11821458	Adapter Plate, Shock/Vibration
6	11821461	Unit Fixture, Rate Tachometer
7	11833718	Rate Tachometer - Vibration Input Drive
8	11823404	Table, Rate of Turn
9	11821668	Tach Generator Drive Fixture - Altitude

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4.6.2 Performance tests.

4.6.2.1 Output voltage linearity. Use item 1 of table VI to verify compliance with the requirements of 3.3.1.

4.6.2.2 Output voltage ripple. Use item 1 of table VI to verify compliance with the requirements of 3.3.2.

4.6.2.3 Rest output. Use item of table VI to verify compliance with the requirements of 3.3.3.

4.6.2.4 Starting torque. Use item 1 of table VI to verify compliance with the requirements of 3.3.4.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 4, 5 and 6 of table VI and subject the assembly to the shock test specified in MIL-STD-81O, method 516.2, procedure I figure 516.2-2. Apply three half since wave shock pulses in each direction along the three axes. Peak amplitude shall be 50 ±5 g with a time duration of 18 + 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.1.2 High intensity. Use items 4, and 6 of table Vi and subject the assembly to the shock test specified in MIL-STD-81O, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock PUISES in each direction along the three axes. Peak amplitude shall be 125.0 ± 12.5 g with a time duration of 0.6 ± 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.2 Vibration. Application of Vibration A and Vibration B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.5.3.2.1 Vibration A. Use items 5, 6, 7 and 8 of table VI and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2, proceure VIII, except that the test level shall be the vibration curve shown in figure 4 and duration of exposure shall be no: less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

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4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 3 of table VI and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of 4.6.2 shall be applied.

4.6.3.4 Low temperature. Use item 3 of table VI and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of 4.6.2 shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 5. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2

4.6.3.6 Humidity. Place the assembly in a humidity chamber and subject it to the humidity test specified in MIL-STD-81O, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the assembly to the performance tests of 4.6.2. After exposure, subject the assembly to the performance tests of 4.6.2.

4.6.3.7 Altitude. Use item 9 of the table VI and subject the assembly to the altitude test specified in MIL-STD-81O, method 500.1, procedure I, except that:

a. First, the assembly in an non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than one hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level , subject the assembly to the performance tests of 4.6.2.

4.6.3.8 Dust. Subject the assembly to the dust test specified in MIL-STD-81O, method 510.1, procedure I. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.9 Salt fog. Subject the assembly to the corrosion test specified in MIL-STD–81O, method 509.1, procedure I. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

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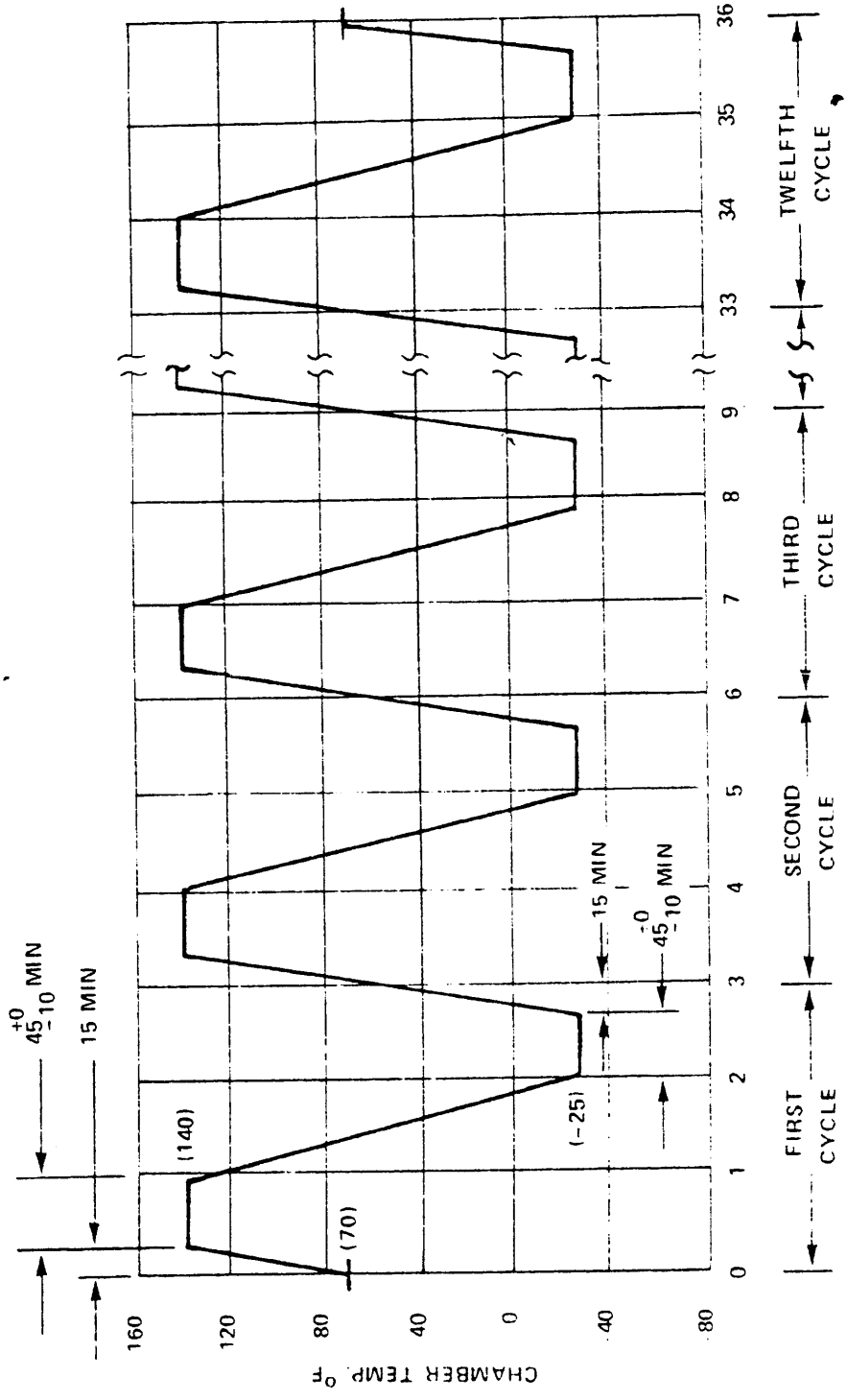


FIGURE 5. Temperature cycling.

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4.6.4 Sealing. Pressurize the assembly with nitrogen to 5.0 ± 0.1 psig, exercise all manual controls and perform either Sealing A or Sealing B leakage tests. After determination of compliance with the sealing requirement, purge the assembly with dry nitrogen at 1 psig for not less than 5 minutes and lower the pressure to 0.0 + 0.2 - 0.0 psig. Nitrogen shall be BB-N-411, type I class 1, grade B with a dewpoint not greater than -25°F.

4.6.4.1 Test selection. Application of Sealing A and Sealing B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.5.4.2 Sealing A. Use item 2 of table VI and subject the assembly to the leakage (immersion) test specified in MIL-STD-810, method 512.1, procedure IV. Immersion time shall be not less than 5 minutes.

4.6.4.3 Sealing B. Use item 2 of table Vl, place the assembly in the following environment for not less than 2 hous and verify that the pressure drop is not greater than 0.2 psig.

a. Temperature	73° + 2.5°F
b. Relative humidity	50 percent ± 5 percent
c. Atmospheric pressure	28.5 + 2.0 - 3.0 in. Hg.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in acccrciance with packaging data sheet P11732700. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. Notes

6.1 Intended use. The assembly is a unit 01 the M21 Ballistic Computer Svsrem.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Special tests and examinations required, the types of inspections required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

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d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-T-48460B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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